

Active learning expedites cross-situational word learning—even under Zipfian  
distributions

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## Abstract

Children learn thousands of words in the first several years of life, inspiring many theoretical and empirical studies of the mechanisms and pace of word learning. Past mathematical analyses have asked how long it should take a child to build an adult-sized vocabulary if words are encountered via random sampling, and concluded that this simple process can succeed in a plausible amount of time—but only under the simplifying assumption that every word is equally likely to be encountered. We show that this optimistic conclusion depends heavily on that assumption: under the more realistic, highly-skewed (Zipfian) word and referent frequency distributions observed in child-directed speech and in real-world scenes, random sampling requires an implausibly large number of learning episodes. We then ask whether a learner (or caregiver) who can influence which words and referents co-occur, rather than simply encountering them at random, can close this gap. Using both mathematical analysis and simulation—including a more realistic incremental learner that does not assume words are learned in a single exposure—we show that such active learners acquire Zipfian vocabularies substantially faster than passive learners exposed to the same random samples, with the size of the advantage depending on how idealized the learning scenario is. These results suggest that the statistical structure that children (or their caregivers) impose on everyday learning situations, rather than sheer exposure alone, may be an important piece of the explanation for how children learn vocabulary as quickly as they do.

*Keywords:* Word learning; Active learning; Zipf’s law; Fast mapping; Linear programming; Word frequency

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### **The remarkable pace of word learning**

Language acquisition proceeds quickly in young children, who within their first year learn the sounds of their language (Kuhl, Williams, Lacerda, Stevens, & Lindblom, 1992), begin segmenting speech (Saffran, Aslin, & Newport, 1996) and even produce their first words (Hulit & Howard, 2002). By 18 months, children can produce on average 50 words and comprehend 100-150 (Hulit & Howard, 2002), with vocabulary size growing to around 1,000 words by age 3 (Hart & Risley, 1995), and reaching roughly 60,000 words by 18 years of age (Bloom, 2000). Under the assumption that each child has 8 hours of word learning opportunity every day, these estimates have the child learning a new word every learning hour for 18 years, on average. This pace seems remarkable, given the storied complexity of the learning environment: a given word could refer to any of infinite possible meanings in any given situation (cf. Blythe, Smith, & Smith, 2016). Although older learners no doubt accelerate word learning as they begin to read, children already show an accelerating pace of learning in the second year of life, growing from an average of 11 words at 12 months to over 300 words by 24 months (Fenson, Marchman, Thal, Dale, & Reznick, 2007; Frank, Braginsky, Yurovsky, & Marchman, 2021). How might young children tackle the daunting problem of mapping words to referents?

Throughout this paper we follow prior theoretical work in analyzing the feasibility of an 18-year, 60,000-word target (Blythe et al., 2016; Blythe, Smith, & Smith, 2010; Vogt, 2012), since this is the estimate against which our results are most directly comparable. It is worth being explicit, however, about which portion of vocabulary growth this target best describes. A commonly-cited developmental trajectory has children recognizing about 50 words at age one, about 1,000 words by age three, and at least 10,000 words by age five (Shipley & McAfee, 2015)—consistent with independent estimates of roughly 10,000 root-word meanings known by early elementary school (Anglin, 1993). Age five is also, for most children, when formal reading instruction

begins, after which written text becomes an increasingly important source of new vocabulary, one governed by a different frequency distribution, a different sampling process, and plausibly different active and passive mechanisms than the largely auditory, caregiver-mediated input we model here. The sampling and learning models we analyze—which assume a single, stationary word frequency distribution encountered through spoken situations—are therefore most directly applicable to this pre-literacy period, during which vocabulary growth is driven overwhelmingly by spoken input rather than text, and less directly applicable to the school-age years during which the bulk of the 60,000-word target is actually reached. We are explicit about this scope limitation up front because it shapes how the 60,000-word, 18-year figures used throughout the paper (following past work) should be read: as a common point of comparison with prior theoretical analyses, not as a claim that our spoken-language sampling models are meant to describe word learning across the full 18-year span. We return to the implications of this scoping for our conclusions in the Discussion.

A variety of heuristics and mechanisms are thought to help learners mitigate referential uncertainty, and empirical studies have shown that young children can take advantage of many of them in the laboratory. For example, children can leverage social cues (Hirsh-Pasek, Golinkoff, & Hollich, 2000), can learn a word from a single informative instance (i.e., *fast-map*; Carey & Bartlett, 1978; Spiegel & Halberda, 2011), tend to assume words refer to whole objects (Macnamara, 1972), and are biased to form mutually exclusive word-referent mappings (Merriman & Bowman, 1989). These heuristics are assumed to aid the process of *cross-situational learning*: acquiring words by accumulating evidence for meanings across multiple contexts—an ability demonstrated in the lab by infants and adults (L. Smith & Yu, 2008; Yu & Smith, 2007). However, determining the relative contributions of these myriad heuristics to naturalistic cross-situational word learning remains a challenge, despite their inclusion in various computational models of word learning, both because such models are primarily applied to experimental settings, and because the complexity and structure of real-world learning environments is largely unknown (though see Clerkin, Hart, Rehg,

Yu, & Smith, 2017; L. B. Smith, Jayaraman, Clerkin, & Yu, 2018; Suanda, Barnhart, Smith, & Yu, 2018).

Moreover, there is disagreement over the basic mechanisms governing word learning: hypothesis-testing models view the problem as one of inference in which the learner may propose and test a single mapping for each word and referent (e.g., Trueswell, Medina, Hafri, & Gleitman, 2013), while associative models gradually accumulate noisy knowledge traces representing graded links between multiple words and referents (e.g., Kachergis, Yu, & Shiffrin, 2012; McMurray, Horst, & Samuelson, 2012). Both types of models can implement fast mapping, mutual exclusivity, and other heuristics, and although they are not always equivalent, it has been shown that these models may often mimic one another’s behavior (Yu & Smith, 2012), making it difficult to choose a general approach to modeling word learning. Nonetheless, it has proven useful to test the overall strength of simple learning mechanisms in idealized simulations of the language learning environment in order to explore the long-term behavior and plausibility of such mechanisms for learning an adult-sized (60,000-word) vocabulary (Blythe et al., 2016, 2010; Vogt, 2012).

**Distributions of words and referents.** Past analyses have generally modeled word learning as a process that randomly samples a single word together with a situation containing a target meaning and, often, several distractor meanings. For example, Blythe et al. (2016, 2010) analyzed an idealized learning model that learns each word on its first sample (i.e., *fast-mapping*, or one-shot learning), in order to estimate the fastest possible learning rate needed to learn a full-sized vocabulary. Because fast-mapping succeeds the instant the target word and its referent co-occur, distractor meanings play no role in this model: the learner simply needs the correct word-referent pair to appear, so the relevant distribution is the frequency with which each word (equivalently, its unique referent, under a mutually-exclusive word-referent mapping) is sampled. Under the simplifying assumption that this word frequency (WF) is uniformly-distributed (Figure 1, top left), Blythe et al. (2010)’s mathematical analysis of the model showed that a cross-situational learner is likely to learn an

adult-sized vocabulary (i.e.,  $\sim 60,000$ ) words after only  $\sim 940,000$  samples, a number of spoken words that is well within the bounds of human life experience, requiring roughly 142 learning instances per day for 18 years. Blythe *et al.*'s analytical estimate has been treated as a theoretical implication that learning via independent fast-mapping of words is sufficiently powerful to be a model of child word learning. Blythe *et al.* (2010) also considered a Zipf-distributed word distribution, finding that it should be slower by a factor that depends on the size of the to-be-learned vocabulary (11.58 for 60,000 words, so 10.9 million).

A separate question is what happens when situations contain multiple candidate meanings whose *identities*, not just the target word's frequency, matter to the learner—for instance, a learner who must eliminate distractors across several exposures rather than fast-mapping in one shot. Analysis of egocentric views from head-mounted cameras on 8- to 10-month-old infants shows a right-skewed frequency distribution of the objects children actually attend to (Clerkin *et al.*, 2017), suggesting that the meanings available in naturalistic learning situations are themselves Zipfian, independent of how often any given word is spoken. Vogt (2012) modeled exactly this scenario: words and their possible (distractor) meanings were both drawn from Zipfian distributions, and because his learner narrows down a word's referent by eliminating distractors across episodes (rather than fast-mapping), a highly-frequent distractor meaning tends to co-occur with the target on every exposure, slowing learning even further. Using Monte Carlo simulation, Vogt (2012) found that learning times increase super-exponentially as a function of the ratio of the context size ( $C$ ) to the number of incidental meanings ( $M$ ), with human-scale learning only plausible for ratios of  $C/M \leq .1$ —i.e., situations in which fewer than a tenth of all possible meanings are plausible candidates. This distractor-elimination slowdown is conceptually distinct from the slowdown we analyze in the next section: ours arises purely from a nonuniform *word* frequency distribution under fast-mapping, where distractor identities are irrelevant by assumption, whereas Vogt's arises from the composition of the distractor set under a different (elimination-based) learning mechanism. We return to the role of situation

composition—which becomes central rather than incidental—when we consider active learners in Section below.

**Active learning.** In stark contrast to these sampling models of word learning, which treat the learner as a passive recipient of externally-determined input, a growing body of work in neuroscience and cognitive science emphasizes that learners actively shape what they attend to and experience, driven by curiosity and the pursuit of informative input (Gottlieb & Oudeyer, 2018). Infants and young children direct their gaze, reach, and requests toward informative objects and events, and structure joint-attention episodes with caregivers in ways that concentrate a single object in view at a time (Yu, Smith, Shen, Pereira, & Smith, 2009), with young children also showing measurable benefits from being given active control over their own learning (Partridge, McGovern, Yung, & Kidd, 2015). Analogous benefits of self-directed sampling have been demonstrated in adult word learners given explicit control over which items to study (Kachergis, Yu, & Shiffrin, 2013; Zettersten & Saffran, 2021). These findings raise the possibility that active selection of learning situations—rather than passively-sampled input alone—may be part of how children (or their caregivers) overcome the slow learning implied by the analysis above.

## The present study

The present study reinspects independent fast-mapping under a more realistic power-law word frequency distribution (WFD) that is characteristic of natural language (Zipf, 1949). Our new mathematical analysis implies that the learning rate of independent fast mapping is quite sensitive to nonuniform distributions of words and referents. Critically, under a Zipfian word frequency distribution, even fast-mapping—the most generous passive learning scheme, since it requires only a single co-occurrence to learn a word—implies an implausibly large number of learning episodes to acquire 60,000 words within 18 years. Intuitively, the difficulty with quickly learning a long-tailed distribution under random sampling is that a learner must wait a very long time to hear the large proportion of low-frequency words. This is a useful diagnostic precisely

because fast-mapping is the best case for passive, independent word learning: if even this idealized scheme implies an implausible number of samples, it suggests that random sampling alone is an incomplete account of how children learn a Zipfian vocabulary, whatever the details of how any individual word is learned. We confirm that simulated learning from the Zipfian WFD in the CHILDES database (MacWhinney & Snow, 1990), a corpus of child-directed speech, is similarly slow, and—going beyond fast-mapping—show with simulation that a more realistic incremental learner, which accumulates evidence for a word-referent mapping across multiple exposures rather than guessing correctly on the first one, shows the same qualitative pattern (Section ).

Given the difficulty of learning naturally-occurring Zipfian WFDs even with the power of fast-mapping, and motivated by the active and curiosity-driven learning literature reviewed above, we explore an extension of the word learning problem in which the learner is not simply a passive recipient of randomly-selected words and referents (as in Blythe et al., 2016, 2010; Vogt, 2012), but instead has some ability to structure the situations from which they will learn: choosing which referents to bring into a situation, from which a caregiver then selects a target referent to name (see Figure 1). Our analysis shows that this form of active learning yields substantial speedups over passive random sampling—in some cases bringing learning times within a plausible range for real-world timescales—even under a long-tailed Zipfian WFD, though the size of the speedup depends on how much of the learning process is idealized (Sections –).

## Methods

The sections that follow analyze several related models, which can be hard to keep straight without a shared vocabulary and a concrete example to anchor the notation. This section lays out both, and previews how the pieces fit together.

### Shared ingredients

Every model in this paper shares the same basic scenario. There are  $W$  words to be learned, and—for simplicity, following Blythe et al. (2010)—exactly  $W$  referents



(meanings), with each word paired with one and only one referent (a *mutually-exclusive* lexicon; we return to this simplifying assumption, including the role of synonymy, in the Discussion). The learner experiences a sequence of *episodes*. In each episode, they encounter a *situation*: a spoken word, together with a set of candidate referents present in the scene, one of which (the *target*) is the word’s true referent and the rest of which (if any) are *distractors*.

Two things can vary independently across the models we analyze:

- **How situations arise.** A *passive* learner simply encounters situations sampled at random according to some external word (and referent) frequency distribution—uniform, in the simplest case, or Zipfian, as in natural language (Zipf, 1949). An *active* learner (or their caregiver) instead has some influence over which referents are brought into a situation—for instance, choosing to play with a toy whose name is not yet known—which in turn changes which words a caregiver is likely to say (Section ).
- **How a situation updates the learner’s knowledge.** A *fast-mapping* learner adopts the word-referent pairing suggested by a single episode immediately and permanently—the most generous possible learning rule, since it requires only one co-occurrence. An *incremental* learner instead accumulates evidence for candidate word-referent pairings across multiple episodes, and only commits to a mapping once the evidence reliably favors it (Section ).

## A running example

Concretely, imagine a toddler learning the names of five toys scattered around a room: a ball, a cup, a dog, a car, and a hat. In one episode, a caregiver says “ball” while the child’s attention includes the ball itself (the target) along with the cup and the dog (two distractors), but not the car or the hat. A fast-mapping learner is simply assumed to correctly identify the ball as “ball”’s referent on this very first exposure, without any account of how the cup and dog come to be ruled out; this is best understood as a mathematical idealization of the fastest any learner could possibly do,

not a mechanistic proposal about how reference is actually resolved (a point on which prior work in this tradition has, fairly, been criticized as underspecified—see Section ). An incremental learner would instead note that “ball” co-occurred with the ball, cup, and dog this time, and would need further episodes—in which “ball” co-occurs with the ball but not consistently with the cup or dog—before confidently learning the mapping. Whether the child hears “ball” at all in a given episode, and which other toys happen to be nearby when they do, depends on whether toys are selected passively (e.g., whatever happens to be within reach) or actively (e.g., the child gravitates toward toys whose names they do not yet know).

## Roadmap

Section analyzes fast-mapping under passive sampling, first for a uniform word frequency distribution (recovering Blythe et al. (2010)’s original analysis), then for Zipfian and other skewed distributions, and finally for the empirical word frequency distribution of the CHILDES corpus of child-directed speech (MacWhinney & Snow, 1990). Section then introduces active learning, first analyzing the idealized case in which the learner has full knowledge of how referents and situations relate, then a more realistic online learner that must estimate this relationship from experience. Finally, Section reports simulations of the incremental (non-fast-mapping) learner introduced above, crossed with passive and active sampling, to test whether the qualitative pattern of results survives when the most idealized assumptions of the earlier sections are relaxed.

## Independent fast-mapping learning

### Uniformly-distributed word frequency

Blythe et al. (2010) proposed a mathematical model of cross-situational word learning which has a closed-form expression under a certain simplification. In their recent study, Blythe et al. (2016) analyzed essentially the same model, slightly modified for analytic convenience. Under this model, let there be  $W$  words and  $O$  referents in the

hypothetical world, and let the number of words be equal to the number of referents,  $W = O$ . Furthermore, let the word-referent mappings be mutually-exclusive: each word corresponds to one (and only one) referent, so there are  $W$  correct word-referent pairs. Without loss of generality, denote the  $W$  pairs by  $1, 2, \dots, W$ , and suppose the  $k^{\text{th}}$  referent is paired with the  $k^{\text{th}}$  word.

Starting with these unknown words and referents, the learner is to infer correct pairs by experiencing *episodes*. In each episode, the learner is exposed to one word  $w_k$ , and  $1 < M < W$  referents, including the target referent  $o_k$ , but without any explicit information about the intended mapping of  $w_k - o_k$ . The simplest scheme is a fast-mapping learning model, which assumes one-shot learning: in one exposure the learner can map a new label to the correct referent.

As fast-mapping learning is the most efficient scheme for independent word learning, it gives a good baseline estimate of the number of samples to learn all the words in a vocabulary. Blythe et al. (2010) model a fast-mapping learner acquiring words independently drawn from a uniform distribution of  $W$  words given in each episode. As the probability of a given word in one episode is  $1/W$ , this is equivalent to the so-called Coupon Collector’s Problem (Blom, Holst, & Sandell, 1994), which asks how many coupons one needs to draw with replacement before having drawn each of  $n$  coupons (words) at least once. In this problem, the expected time  $T$  to finish sampling all the words is

$$E[T] = W \sum_{i=1}^W 1/i \approx W \log W. \quad (1)$$

For an adult-sized vocabulary  $W = 60,000$ , the estimated number of samples needed would be  $T = 660,126$ . This estimate is comparable with the “reasonable” number of samples justified by (Blythe et al., 2010) which individual children can be exposed to for their 18 years of life.

### Number of samples for a general WFD

We now extend this analysis of fast-mapping to the more realistic case of non-uniform WFDs. This analysis will reveal that the estimate based on Equation (1)

by Blythe et al. is overly-optimistic: learning of non-uniform distributions is generally much slower.

Suppose a set of  $W$  words in which each word  $1, \dots, W$  is drawn from the distribution  $p = (p_1, \dots, p_W)$ . Let us write the number of episodes by  $T$  which is required to have the proportion of children who finished learning all the words is  $(1 - \epsilon)$  for  $0 \leq \epsilon \leq 1$ . For the uniform distribution,  $p_i = 1/W$  for every  $i = 1, \dots, W$ , the number of samples is

$$T = f_{W,\epsilon}(1/W), \quad (2)$$

where

$$f_{W,\epsilon}(x) := \frac{\log(1 - (1 - \epsilon)^{1/W})}{\log(1 - x)}.$$

Intuitively,  $f_{W,\epsilon}(x)$  is just the coupon-collector calculation behind Equation 1, generalized to ask: how many samples are needed before a word with sampling probability  $x$  has appeared at least once, for all but a proportion  $\epsilon$  of learners? Setting  $\epsilon = 1/2$  in (2), we obtain the median of  $T$ ,  $f_{W,1/2}(1/W)$ , that is comparable with the mean of  $T$  in (1).

Unlike (2) for the uniform distribution, the number of samples  $T$  for an arbitrary WFD in general does not have a closed-form solution. Thus, instead of the exact expression of  $T$ , we consider the upper and lower bound for  $T$  (see Appendix 1 for the derivation). For the general WFD  $p = (p_1, \dots, p_W)$ , we have inequality

$$f_{W,\epsilon}(\max p) \leq T \leq f_{W,\epsilon}(\min p) := T_+. \quad (3)$$

As we are interested in the worst possible estimate of  $T$ , this inequality states that the upper bound  $T_+$  of  $T$  is characterized with the probability to sample the least frequent word  $\min p$ . In words: among all  $W$  words, it is the single rarest one—with probability  $\min p$ —that determines how long a learner must wait to complete the vocabulary, since every more-frequent word will typically already have been sampled well before it. This is why nonuniform word frequency distributions are so much harder to learn from than uniform ones: a Zipfian distribution concentrates most of its probability mass on a handful of frequent words, leaving very little for the long tail of

rare ones, so its  $\min p$  is far smaller than the  $1/W$  every word gets under a uniform distribution—and, by Equation (3), a smaller  $\min p$  directly translates into a much larger  $T_+$ .

### Impact of non-uniform word frequency

This extended mathematical analysis implies that the uniform distribution  $q = (1/W, \dots, 1/W)$  of words gives the minimal possible upper bound  $T_+$  among any WFD of  $W$  words, as any distribution  $\min p$  of  $W$  words holds  $\min p \leq \min q$ . Therefore, the expectation of  $T$  in the form of (1) with uniformly-distributed words is the most optimistic, and thus likely underestimates the number of episodes required for learning a realistic WFD.

For example, consider the case that the  $W$  words are Zipf-distributed with exponent  $a \geq 0$  defined by

$$p = (1^{-a}/H_{W,a}, 2^{-a}/H_{W,a}, \dots, W^{-a}/H_{W,a})$$

where  $H_{W,a}$  is the generalized harmonic number  $H_{W,a} = \sum_{i=1}^W i^{-a}$ . With  $a = 0$ , this reduces to a uniform distribution. The larger the exponent  $a$  is, the minimal probability  $\min p$  is smaller. In the case of the “standard” Zipfian distribution  $a = 1$ , the minimal probability is  $\min p \approx 1.44 \times 10^{-6}$ , and the upper bound  $T_+$  is  $1.08 \times 10^7$  for  $\epsilon = 0.01$ . This estimate means that learning a Zipfian WFD requires 16.4 times as many samples as learning a uniform WFD. This implies 206 independent episodes exposed to a word learner per hour—three episodes every minute—assuming 8 hours of learning everyday of 18 years of life. This number of independent episodes does not seem reasonable with respect to the ordinary life of children in any culture.

### Sensitivity to non-uniformity of word frequency distribution

To analyze how sensitive the number of samples  $T$  is to non-uniformity, we analyze Zipfian WFDs with different exponents. With some calculation (see Appendix 2), we find the upper bound  $T_+$  (Equation 3) as a function of the exponent parameter  $a$

holds the bound

$$\frac{\partial^2 \log T_+}{\partial a^2} \geq 0.$$

This implies the  $T_+$  is a super-exponential monotone function of the exponent  $a$ . It is also numerically confirmed in Figure 2, which shows the number of episodes as a function of the WFD’s exponent for  $W = 10000, 60000$ . In this plot,  $a = 0$  shows an estimate for the uniform distribution, and  $a = 1$  shows that of the standard Zipfian distribution. It is striking that even fast-mapping—the most efficient learning—can be quite slow (exponentially as a function of  $a$ ) for distributions that have some low-probability items.

### Empirical dataset

Given these theoretical implications, we analyze an empirical WF distribution of child-directed speech in the CHILDES corpus (MacWhinney & Snow, 1990). Figure 3 shows a representative word distribution of 51,446 words aggregated over 4,163 transcripts from CHILDES (retrieved December 2007). The minimal word probability was  $1.089 \times 10^{-7}$ , which gives the upper bound

$$T_+ = f_{51446,0.01}(1.089 \times 10^{-7}) = 1.420 \times 10^8 \text{ or the median estimate}$$

$$T_+ = f_{51446,0.5}(1.089 \times 10^{-7}) = 1.030 \times 10^8. \text{ These estimates of required samples, being an order of magnitude larger than the optimistic theoretical estimate, suggest that this empirical Zipfian WF distribution would be difficult to learn.}$$

A caveat about this estimate is worth making explicit, since Equation (3)’s bound  $T_+$  is driven entirely by the single least-frequent word in the corpus. This frequency distribution is aggregated across 4,163 separate transcripts, each contributed by a different family; the very lowest-frequency words in Figure 3 may therefore include words idiosyncratic to a single household (e.g., a pet’s or sibling’s name) rather than words that are genuinely rare in any individual child’s actual input. If so, the corpus-wide minimum overstates how rare the rarest word in an individual child’s own environment really is, and our  $T_+$  estimate would overstate that child’s learning time. Working in the opposite direction, a corpus of this size may also under-sample genuinely

rare but widely-shared words, pulling the estimated minimum probability—and hence  $T_+$ —the other way. Because we cannot fully adjudicate between these two sources of bias without frequency counts drawn from individual children’s longitudinal input rather than pooled across families, we treat the CHILDES-based estimate reported here as illustrating the same order-of-magnitude difficulty implied by our idealized Zipfian analysis above, rather than as a precise quantitative bound in its own right.

### Active learning: Choosing situations

#### Formulation

The mathematical analysis above suggests that even the power of fast-mapping may not be efficient enough to learn Zipfian WFDs, raising the question of what mechanisms enable children to learn the nonuniform distributions they experience. We explore one way to reconcile this discrepancy by adjusting the assumption about how about meanings are sampled in each situation. In the analysis so far, the learner is assumed to be *passive*, having no choice but to observe and learn from a given situation: a randomly-sampled set of objects, of which a (random) subset are labeled with words. The simplifying assumption of a passive learner likely underestimates real learners (and caregivers), who have some control over which situations are experienced. Here, we consider a type of active learner who is able to *actively* choose the situations from which she learns words. Returning to the toy example from the Methods section: rather than being handed a random toy, an active learner (or their caregiver) can choose to set up a particular scene—say, gathering the ball, the car, and the hat, rather than the ball, the cup, and the dog—and different scenes make different words more or less likely to be spoken by a caregiver.

Suppose that there are  $N$  word-referent pairs and  $M$  possible situations (e.g., scenes the learner could construct), and that the conditional probability to observe the  $i^{\text{th}}$  word-referent pair is  $p_{ij}$  given the  $j^{\text{th}}$  situation. Assume the active learner can choose the next learning situation from the given  $M$  situations. Suppose that the active learner chooses the  $j^{\text{th}}$  situation with probability  $q_j$ . Denote the  $N \times M$  matrix of conditional

probabilities by  $P = \{p_{ij}\}_{ij}$  and the  $N \times 1$  vector of the situation choice probabilities by  $q = (q_1, q_2, \dots, q_M)^T$ . Thus, the marginal probability of word-referent pairs is given by the vector  $Pq \in \mathbb{R}^N$ . Concretely, if the learner always constructed the same scene  $j$ , the words they would hear would simply follow that scene's column of  $P$ ; by mixing across scenes according to  $q$ , the learner reshapes the overall word frequency distribution they experience,  $Pq$ , relative to whatever distribution a passive observer of randomly-constructed scenes would encounter. According to the analysis in the previous section, the minimal probability of referents (or words) determines the number of samples required to complete the word learning, meaning the best choice for the active learner is given by the choice probability <sup>1</sup>

$$\hat{q} = \arg \max_q \min(Pq).$$

In words,  $\hat{q}$  is whatever mixture of situations makes the *least*-likely word-referent pair as likely as possible—maximizing the minimum entry of  $Pq$ —because it is exactly this minimum probability that, per Section , determines how long the learner must wait to complete the vocabulary. Framed this way, active learning as formalized here does not change how quickly any individual word is learned once encountered; rather, it reshapes which words the learner is likely to encounter in the first place, spreading exposure more evenly across the vocabulary than random sampling of situations would. This minimal probability,  $\min(P\hat{q})$ , gives the theoretical upper bound for the minimal number of samples

$$T_+ = f_{W,\epsilon}(\min(P\hat{q})).$$

This is an upper bound with the known  $P$ , because the probability matrix  $P$  is not known before empirical learning, and the active learner also needs to estimate  $P$  from the sample.

For a given matrix  $P$ , the optimal  $\hat{q}$  can be computed by the iterated linear programming algorithm (see Appendix 3 for details).

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<sup>1</sup> In the context of game theory, this optimal  $\hat{q}$  corresponds to the max-min strategy (or Nash equilibrium) for the two-player zero-sum game defined by the payoff matrix  $P$  (see for instance Maschler, Eilon, and Shmuel (2013)).



As a baseline for the passive learner, we consider the average  $\min(Pq)$  with the uniform distribution over the vector  $q$ —i.e., a learner (or caregiver) who constructs scenes without regard to which words the learner has already mapped to a referent, the counterpart of the randomly-sampled situations analyzed in the previous section, now expressed in terms of situations rather than words directly. This baseline’s lower bound is given by Jensen’s inequality

$$\int_{q \in \mathbb{S}^N} \min(Pq)(N-1)! dq \geq \min(P\mathbf{1}_N/N),$$

where the integral is taken over the  $N-1$  dimensional unit simplex  $q \in \mathbb{S}^N$ . For a sufficiently small  $x \ll 1$  and  $y \ll 1$ ,  $f_{W,\epsilon}(x)/f_{W,\epsilon}(y) \approx y/x$ . Thus, the rate

$$R = \min(P\hat{q})/\min(P\mathbf{1}_N/N)$$

gives a good estimate for the rate of efficiency, by which active learning with the optimal probability  $\hat{q}$  is  $R$  times faster on average than passive learning with a uniformly-randomly chosen probability vector  $q$ .

### Empirical evaluation

To evaluate the potential impact of active learning, we study the SUN database Xiao, Hays, Ehinger, Oliva, and Torralba (2010) as an empirical estimate of object distributions in a collection of real-life scenes. The SUN database (retrieved September 25, 2016) has  $N = 3,458$  objects and  $M = 1,111$  representative indoor and outdoor scenes in 131,067 images. This data gives an estimate of the  $N \times M$  matrix  $P$  in which each column is the conditional probability of the objects given each scene. With uniform scene choice probability  $q = \mathbf{1}_M/M$ , the  $\min(Pq)$  is  $8.30 \times 10^{-9}$ . Meanwhile, with the optimal  $\hat{q}$ , the  $\min(P\hat{q})$  is  $1.95 \times 10^{-6}$ , indicating active learning is approximately  $R = 235.3$  times faster than the baseline passive learning. The marginal probability distributions of objects for the baseline and optimal  $q$  are shown in Figure 4. The difference between the two marginal distributions is visible at their tails: the tail of the uniform  $q$  decreases like an exponential function, but that of the optimal  $\hat{q}$  decreases as a power function (linear in the log-log plot).

This tail comparison also clarifies an important limitation of the comparison: the word-referent distribution realized under uniform scene selection in this dataset is not itself Zipfian, and its exponential tail is in fact *less* heavy—and thus easier to learn from—than the Zipfian distribution assumed as the passive baseline in Section . The baseline passive learner analyzed here is therefore the passive learner implied by this particular scene database under uniform scene selection, not a learner facing the specific Zipfian word frequency distribution analyzed in Part 1 of this paper; the two are related but not identical comparisons. If anything, this suggests the  $235\times$  estimate is conservative relative to a passive learner constrained to face a genuinely Zipfian marginal distribution matched to Section ’s assumptions, since the exponential-tailed baseline realized here should already be somewhat easier to learn from than a Zipfian one. Constructing  $P$  so that its uniform- $q$  marginal is constrained to exactly reproduce a specified Zipfian distribution—allowing a matched, apples-to-apples comparison with Part 1—is a natural target for future work with richer scene databases. This empirical evaluation suggests that this form of active learning can make fast-mapping orders of magnitude more efficient.

### Online active learning with an unknown $P$

The above measure of active learning’s efficiency is based on the optimal  $\hat{q}$  with full knowledge of  $P$ , which gives an optimistic estimate of learning time since the matrix  $P$  is unlikely to be fully known to the learner. Thus, we perform a Monte Carlo simulation to quantify the efficiency of an *online* active learner who gradually learns  $P$  and estimates  $q$  based on the sample estimate of  $P$ . If this online active learner is comparable to the optimal active learner with  $\hat{q}$ , we can treat the performance analysis of the optimal active learner above (a few orders more efficient) as holding for the online active learner. For this purpose, we generated a  $N \times M$  matrix  $P$  with  $N = 1000, M = 100$ , which has the elements in each column are Zipfian probabilities  $P_{\pi(i)} \propto i^{-a}$  with random coefficients  $a \in [1, 1.5]$ , where  $\pi : \{1, \dots, N\} \mapsto \{1, \dots, N\}$  is a random permutation. The online active learner has uniform choice probability

$q_1 = \mathbf{1}_N/N$ . For  $k^{\text{th}}$  batch of 1000 steps, the online learner samples the objects according to the probability  $Pq_k$ , and constructs the sample probability matrix  $\hat{P}_k$  according to sample frequency. After the  $k^{\text{th}}$  sampling step, the online learner estimates  $q_k := \arg \max_q \min(\hat{P}_k q)$ . In each run of this procedure, we repeat up to  $100 \times 1000$  samples, and obtain one sample for the number of required samples to finish learning names for all of the 1000 referents—the approximate vocabulary size of a 3-year-old.

With 100 runs, we obtain the Monte Carlo estimate of the online learner shown in Figure 5. Figure 5 shows the sample probability distribution of the number of required samples in the Monte Carlo simulation (circles: histogram, line: smoothed estimate), and its comparable median estimate for the optimal learner (green vertical line) and the passive learner with the uniform  $q$  (red vertical line). This simulation result shows that the online learner is, on average, as fast as an optimal learner, and is almost always faster than the passive learner, which requires twice as many samples.

Beyond the case study reported above, we also provide a theoretical analysis of why such online learning is possible in Appendix 4.

### Incremental active learning

The analyses above—both the fast-mapping bound of Section and the idealized and online active learners of Section—all assume that a word is learned the moment it co-occurs with its target referent. This lets us derive clean analytical results, but it sidesteps a concern raised about this modeling tradition more broadly: that conclusions about learning speed may reflect the *sampling* process (how often the right word-referent pair happens to co-occur) rather than anything about *learning* itself, since treating learning as instantaneous leaves no room for the two to be distinguished. To test whether our qualitative conclusions—that passive sampling from a Zipfian distribution is implausibly slow, and that active learners can close much of this gap—survive when this instantaneous-learning idealization is relaxed, we simulated a learner who accumulates evidence across multiple exposures rather than fast-mapping. This simulation leaves a second idealization untouched: as in all the models in this paper, words are still learned

fully independently of one another, with no transfer of information from an already-learned word (e.g., “dog”) to a related but distinct one (e.g., “cat”). We return to this remaining simplification, and its likely direction of bias, in the Discussion.

### **An incremental, mutual-exclusivity learner**

Following Vogt (2012) and Siskind (1996), we implemented a learner who tracks, for each word, the set of referents it remains consistent with across all past exposures. In each episode, the learner observes a target word alongside  $C - 1$  distractor referents (as in Section ); rather than immediately committing to a mapping, the learner eliminates from the word’s candidate set any referent that was absent from the current situation. A word is considered learned once its candidate set has been narrowed to a single referent—i.e., once enough situations have ruled out every other possibility. This is a strictly harder learning problem than fast-mapping: a single co-occurrence is no longer sufficient, and how quickly a word is learned now depends on *which* distractors happen to accompany it, not only on how often it is sampled.

We crossed this incremental learner with the same two dimensions of active control introduced above, but now made concrete for both the target word and the distractors that accompany it, following the toy example from the Methods section:

- **Target selection:** a *passive* learner encounters a target word/referent sampled according to the (uniform or Zipfian) word frequency distribution; an *active* learner instead preferentially selects situations built around a referent whose word is not yet known (e.g., choosing to play with the car rather than the already-learned ball).
- **Distractor (context) selection:** under *random* context selection, the  $C - 1$  distractors accompanying the target are drawn according to the word frequency distribution, as in Section ; under *familiar* context selection, distractors are drawn preferentially from referents whose candidate sets have already been narrowed down (i.e., toys the learner already knows a great deal about), which should make it easier to eliminate them as competitors for the target’s word. This mechanism

formalizes a disambiguation strategy with direct developmental support: children as young as two systematically exclude already-known objects as candidate referents for a novel word, and do so more reliably the better they already know those objects (Grassmann, Schulze, & Tomasello, 2015)—precisely the informal mutual-exclusivity intuition our elimination rule implements. It is also consistent with computational and observational evidence that non-co-occurrence information—including which referents have recently been discussed or attended to, and not just which are visually present—measurably improves reference resolution in cross-situational word learning (Frank, Goodman, & Tenenbaum, 2009; Frank, Tenenbaum, & Fernald, 2013).

This  $2 \times 2$  design (active/passive target selection  $\times$  random/familiar context selection), crossed with uniform and Zipfian word frequency distributions, lets us ask which forms of active control matter, and whether their benefits depend on the frequency distribution being learned—all without assuming fast-mapping.

## Simulation design

We simulated vocabularies of  $M = 1000$  and  $M = 2000$  referents, with Zipfian word frequencies following the Zipf-Mandelbrot form  $p_r \propto (r + 2.7)^{-1}$  over frequency rank  $r$  (Mandelbrot, 1953). Context sizes  $C$  ranged from 5 to 200, always satisfying  $C \leq 0.1M$  per Vogt (2012)’s finding that learning becomes intractable outside this range. For each combination of  $M$ ,  $C$ , target selection (active/passive), context selection (random/familiar), and word frequency distribution (uniform/Zipfian), we ran 15–100 replications and recorded the number of episodes needed to learn 99% of the vocabulary.<sup>2</sup>

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<sup>2</sup> Simulation code and raw results are available in the project repository

(`corpusXSL_eliminitive_active_parallel.R`, `activeXSL_simulations.Rmd`).

## Results

Figure 6 shows the mean number of episodes needed to learn 99% of a  $M = 1000$ -word vocabulary, for the subset of context sizes ( $C = 10, 100$ ) simulated under all four combinations of target and context selection. Under a uniform WFD, active target selection alone yields a modest speedup over passive sampling (2,894 vs. 8,340 episodes, a 2.9x reduction), and familiar-context selection adds essentially nothing further—unsurprising, since eliminating distractors is not especially hard when every referent is equally frequent to begin with.

Under a Zipfian WFD, the pattern is very different, and directly addresses the concern that our earlier fast-mapping results might not generalize to a learner that must accumulate evidence over time. Fully passive learning (random target and context selection) requires an average of 5,893,267 episodes—consistent with the implausibly slow learning implied by the fast-mapping analysis of Section , now obtained from a learner that never fast-maps at all. Active target selection alone cuts this to 357,958 episodes, a 16.5x speedup, comparable in order of magnitude to the online active learner of Section . Familiar-context selection alone (with passive target selection) is even more effective, reducing learning time to 56,232 episodes (a 104.8x speedup)—because once a few referents have been narrowed down, using them as distractors makes every subsequent elimination nearly free. Combining active target selection with familiar-context selection reduces learning time to 5,809 episodes on average, barely above the  $M = 1000$  lower bound of one episode per word, and over 1,000 times faster than fully passive learning. In other words, once the learner is allowed to shape not only *which* word they hear next, but also *what else is in the scene* when they hear it, an otherwise implausibly slow Zipfian learning problem becomes nearly as fast as the uniform case.

These results reinforce the paper’s central conclusions using a learner that makes neither of the two idealizations most often raised against fast-mapping models: words are not learned on a single exposure, and the number of samples required is not simply a proxy for how often a word is spoken. They also reveal that the two forms of active control we distinguish—choosing what to talk about, and choosing what else is nearby

when talking about it—are not interchangeable: under a Zipfian distribution, controlling the distractor context turns out to matter at least as much as controlling the target word itself, a distinction that the idealized analysis of Section does not draw out, since it collapses both into a single choice of situation  $j$ .

## Discussion

This study reinspected the claim by Blythe et al. (2010) that independently fast-mapping word meanings is efficient enough to learn a full adult lexicon in 18 years, extending the analysis to arbitrary (not only uniform or Zipfian) word frequency distributions, and complementing Vogt (2012)’s related but distinct analysis of Zipfian distractor sets under a different, elimination-based learning mechanism (see Section ). We offer a first closed-form analytical bound on the learning time of an entire lexicon for an arbitrary word frequency distribution, consistent with the general intuition behind Vogt (2012)’s simulation results that learning time is governed by the probability of sampling the single least-frequent word. Our new analysis showed that even fast-mapping—the most generous passive, independent learning scheme—implies an implausibly large number of samples for word distributions with a long tail of rarely-sampled words, including the Zipf distribution, and simulations of a more realistic incremental learner showed the same qualitative pattern (Section ).

Given that our analysis implied learning would be too slow under realistic distributions, we considered a more efficient learning scheme, in which the learner can choose preferred learning situations by composing the contexts from which words are drawn. This type of active control over situations seems natural with respect to general observations of children’s behavior (Yu et al., 2009), and has been shown to benefit adult word learners (Kachergis et al., 2013), but has not to our knowledge previously been subjected to theoretical analysis. We quantify and evaluate the effect of this type of self-directed learning in word learning using a modeling framework with few assumptions (Blythe et al., 2016, 2010; Vogt, 2012). As the least probable word in the distribution determines learning efficiency, we analyzed the effect of active choice on the

situations maximizing this key parameter. Analyzing an empirical dataset of labeled referents in a set of natural scenes, we found that the expected learning time of an idealized active learner—one with full advance knowledge of how referents co-occur with scenes, who still fast-maps each word—is over two hundred times (235x) more efficient than a passive learner sampling scenes uniformly at random. Two further analyses show that this large speedup is best treated as an upper bound rather than a representative estimate, for two largely independent reasons. First, an active learner who must instead estimate this co-occurrence structure online from experience, rather than know it in advance, but who still fast-maps, requires only about half as many samples as its passive counterpart in our simulated setting (Section )—a much smaller advantage than the idealized case, since the learner spends part of its budget discovering  $P$  rather than exploiting it. Second, and separately, a learner who cannot fast-map at all, but must instead accumulate evidence for word-referent mappings across multiple exposures, shows a roughly 16-fold advantage for active target selection alone under a Zipfian distribution (Section )—and a more than 1,000-fold advantage once the learner can also shape which distractors accompany the target word, rather than only which word is targeted. Taken together, these estimates span roughly two orders of magnitude, from about 2x to over 1,000x, depending on which idealizations are relaxed: how much the learner already knows about its environment (a known vs. estimated  $P$ ), whether learning requires one exposure or several, and whether active control extends only to the target word or also to its surrounding context. What remains stable across this range is the direction and rough scale of the effect, not its exact size: some form of active control over learning situations meaningfully speeds up vocabulary acquisition under Zipfian word and referent distributions, and does so far more than under a uniform distribution, where active selection helps only modestly (Section ). Together, these results suggest that active choice in word learning is one plausible contributor to how children learn Zipfian vocabularies within realistic timeframes—not that passive learning plays no role, but that it is unlikely to be the whole story.

This paper analyzed both an idealized fast-mapping learner and a more realistic



incremental learner in order to characterize the effects of nonuniform word frequency distributions, with and without active learning. We expect that other word learning mechanisms suffer to the same extent from nonuniform distributions in the environment, and future work can use the same analytic techniques presented here to confirm that. Moreover, we expect that the active learning mechanisms presented are generalizable to other models, and will similarly alleviate the difficulty of learning from naturalistic nonuniform distributions. Although the optimal active learning algorithm we have considered may not be cognitively plausible (and thus may predict faster learning than young children show), in conjunction with the fast-mapping mechanism it offers an upper-bound on the speed of active word learning, and future work should consider suboptimal active learning mechanisms that are more cognitively plausible. The cognitive plausibility of the specific mechanisms we simulate is a fair question to press on, and the evidence is genuinely mixed rather than uniformly supportive. On one hand, infants preferentially attend to stimuli of intermediate, rather than minimal or maximal, predictability or complexity (the “Goldilocks effect”; Kidd, Piantadosi, & Aslin, 2012), consistent with our active learner’s preference for referents whose words are not yet known but which are otherwise unremarkable; and self-directed, environment-adapted information search is documented in children from around age three onward (Ruggeri, 2022), well within the age range relevant to our pre-literacy scope (see the Introduction). On the other hand, at least one recent, well-controlled study found no advantage of active over passive information selection for infants’ retention of novel word-object associations in a gaze-contingent paradigm (Bazhydai, Wong, Altmann, Jones, & Westermann, 2026), a direct empirical test of a claim close to the one we model computationally. We do not think this null result undermines our central contribution—which is a computational characterization of how large an advantage active control *could* confer under Zipfian distributions, not a claim that children realize this advantage in full—but it is a reason for caution in treating our active-learning estimates as a description of typical development, rather than as an upper bound that motivates looking for partial, more constrained forms of active

control. Our analysis also rests on simplifying assumptions common to this modeling tradition (Blythe et al., 2016, 2010; Vogt, 2012) that are worth making explicit. First, we assumed a mutually-exclusive lexicon in which each word maps to exactly one referent; in reality many words have synonyms, which may ease learning somewhat (evidence for one member of a synonym set can be informative about related words), but does not change our basic conclusion, since each synonym must still be individually sampled and learned under both the fast-mapping and incremental models we analyze. Second, following Bloom (2000) we targeted a vocabulary of roughly 60,000 words, whereas more recent estimates place adult vocabulary size somewhat lower, around 40,000 word families (Brysbaert, Stevens, Mander, & Keuleers, 2016). More importantly than the exact target size is whether the child must learn a specific, enumerated set of  $W$  words—for instance, the  $W$  least-frequent words in the language, an especially hard case since low-frequency words are by definition rarely sampled—or merely *some* sufficiently large subset of the words available in the environment, of the child’s choosing. The latter framing is substantially easier, but only because it lets the learner restrict attention to the  $W$  *most frequent* available words rather than an arbitrary or unlucky selection; because the most frequent fraction of a Zipfian distribution is itself Zipfian, this restricted problem is reachable in substantially fewer samples than learning the full tail. It is worth being precise about what this reframing does and does not buy the passive learner: it does not, on its own, rescue passive learning from our conclusions, because a passive learner sampling situations at random has no way to arrange for only frequent words to be the ones that get learned—selectively attending to and prioritizing more frequent, salient words is itself a form of active selection, of the same general kind we analyze in Sections –, rather than a property of passive sampling per se. Our qualitative conclusions—that passive fast-mapping is implausibly slow under Zipfian sampling, and that actively structuring learning situations substantially narrows this gap—do not depend on which of these framings, or which exact target vocabulary size, is assumed; only the specific numerical estimates (e.g., Equation 1 and Figure 2) do. Third, as flagged in the Introduction, the

spoken, caregiver-mediated sampling process we model is most directly applicable to the pre-literacy period—during which children are estimated to reach roughly 10,000 words (Anglin, 1993; Shipley & McAfee, 2015), an order of magnitude short of the full 60,000-word target—rather than to the school-age years during which reading and explicit instruction plausibly introduce different sampling and learning mechanisms altogether. We retain the 60,000-word, 18-year figures throughout because they are the benchmark against which Blythe et al. (2016, 2010); Vogt (2012) is most directly comparable. Applying Equation (3) to the more conservative pre-literacy target directly (a Zipfian vocabulary of  $W = 10,000$ ,  $\epsilon = 0.01$ ) gives  $T_+ \approx 1.35 \times 10^6$ , or about 93 episodes per hour over 8 hours a day for 5 years—a somewhat lower bar than the roughly 206 episodes per hour implied by the 60,000-word, 18-year target, since the smaller vocabulary’s tail is less extreme even though the available time is much shorter. This means our qualitative conclusion—that Zipfian sampling is far harder than uniform sampling, and that active learning meaningfully closes the gap—is what should be read against the pre-literacy trajectory, not the specific 206-episodes-per-hour figure, which is somewhat pessimistic if the true target for this period is closer to 10,000 than 60,000 words.

Note that the general scene-selection algorithm analyzed here could be implemented by the child, the caregiver, or both as they jointly choose (or reject) referents and locales. Indeed, observations of parent-child free-play sessions have found that both parents and children often construct scenes with a single object dominating the child’s visual field (Yu et al., 2009), with manual manipulations of both the parent and the child contributing to the visual clarity of the objects (Suanda et al., 2018). Data such as these could be used to test the efficacy of various active learning mechanisms against real-world data, providing additional insights into how restructuring a child’s environment might improve word learning. Applying mechanisms of active learning to the rich and changing input will allow us to test the idea that infants create a curriculum for statistical learning (L. B. Smith et al., 2018).

## Acknowledgment

## References

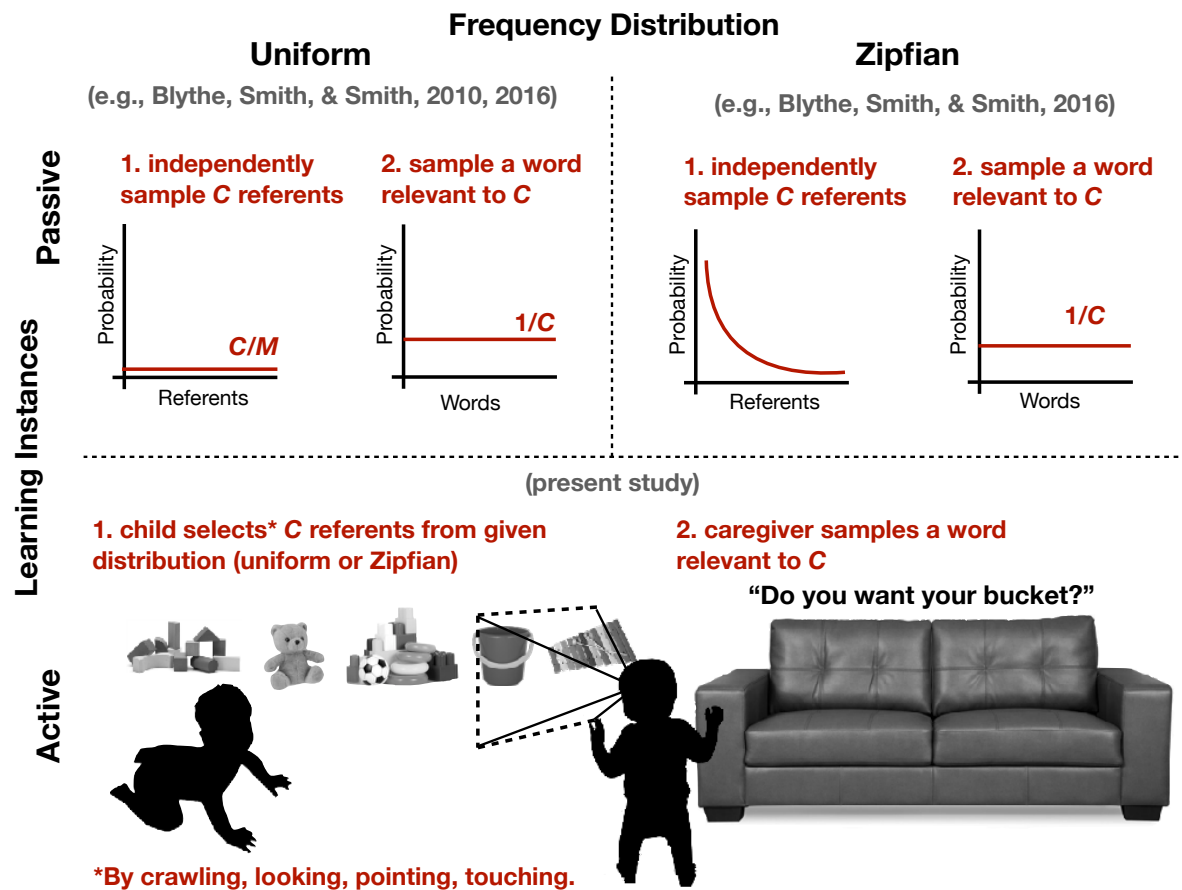
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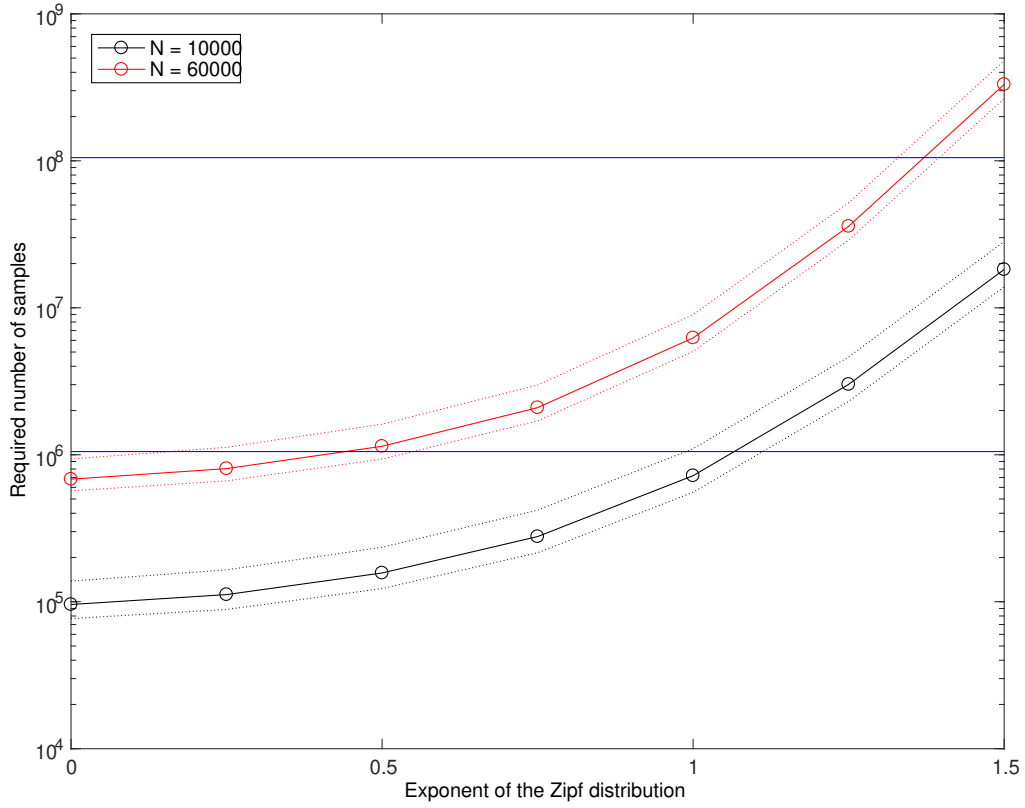
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*Figure 1.* Previous research considered passive learning from independent random samples from uniform (top left) or Zipfian (top right) word and referent (i.e., meanings  $M$ ) frequency distributions. The present work shows that passive learning from a Zipfian word and referent distribution requires an implausibly large number of samples, and explores active learning—a general scheme by which the infant (or caregiver) chooses the situation (i.e., context  $C$ ) from which they will learn (bottom)—as a candidate source of the additional speed children’s learning requires.



*Figure 2.* The required number of samples ( $M$ ) from a generalized Zipf distribution ( $p_k = k^{-a} / \sum_{k=1}^N k^{-a}$ ) needed for 1%, 50%, and 99% of learners ( $\epsilon = [.99, .01]$ ; dotted lines,  $\epsilon = 0.5$ ; solid line) to reach a vocabulary size of  $N = 10000$  or  $N = 60000$ , for varying values of the distribution exponent  $a = 0, \dots, 1.5$ . A value of  $a = 0$  corresponds to a uniform distribution. The number of samples is numerically calculated by the root of Equation (4) in Appendix 1.

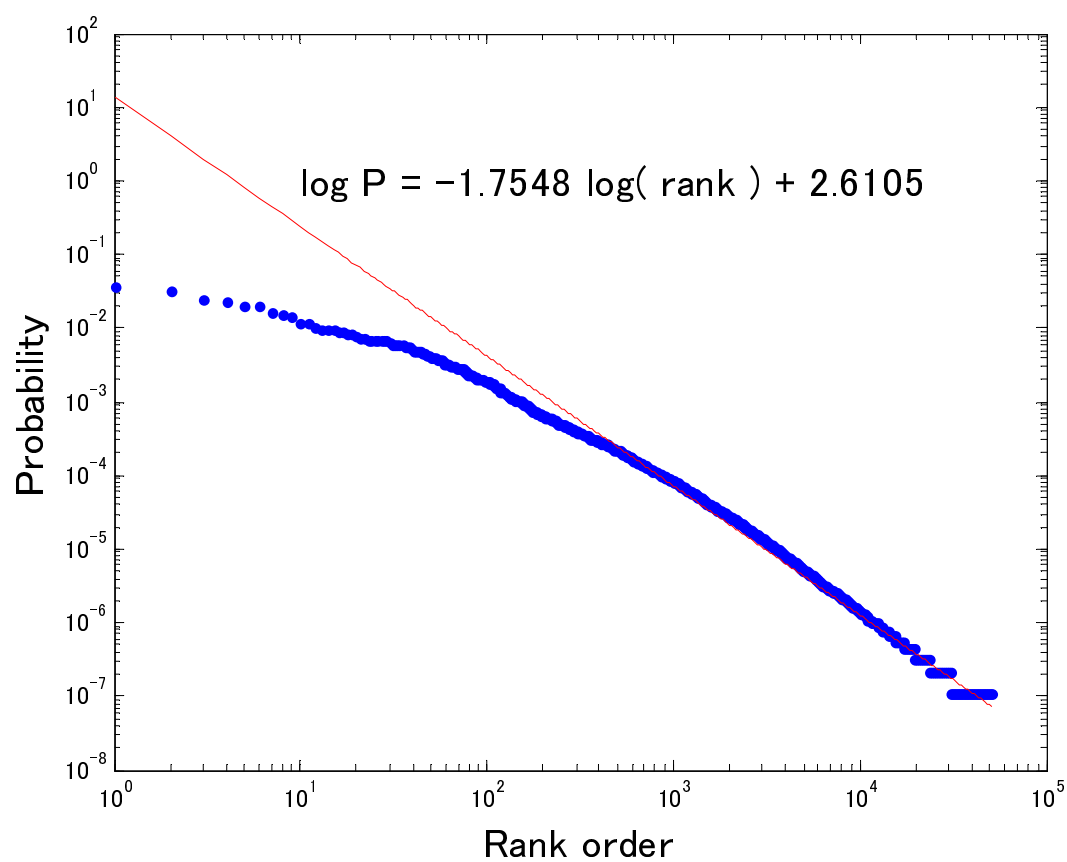


Figure 3. Word frequency in the CHILDES corpus transcripts.

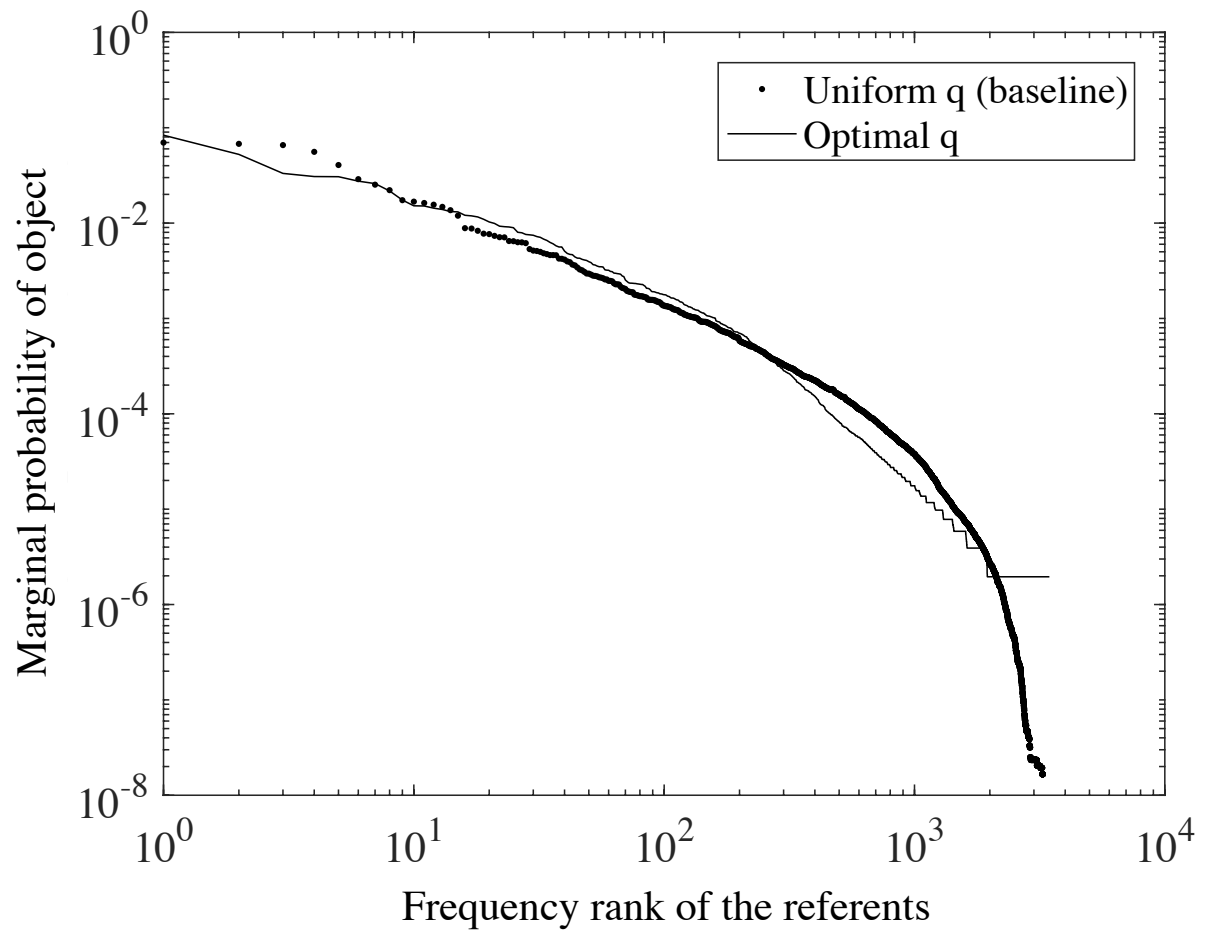


Figure 4. The marginal probability of referents for the optimal  $\hat{q}$  (line) and its baseline (dots).

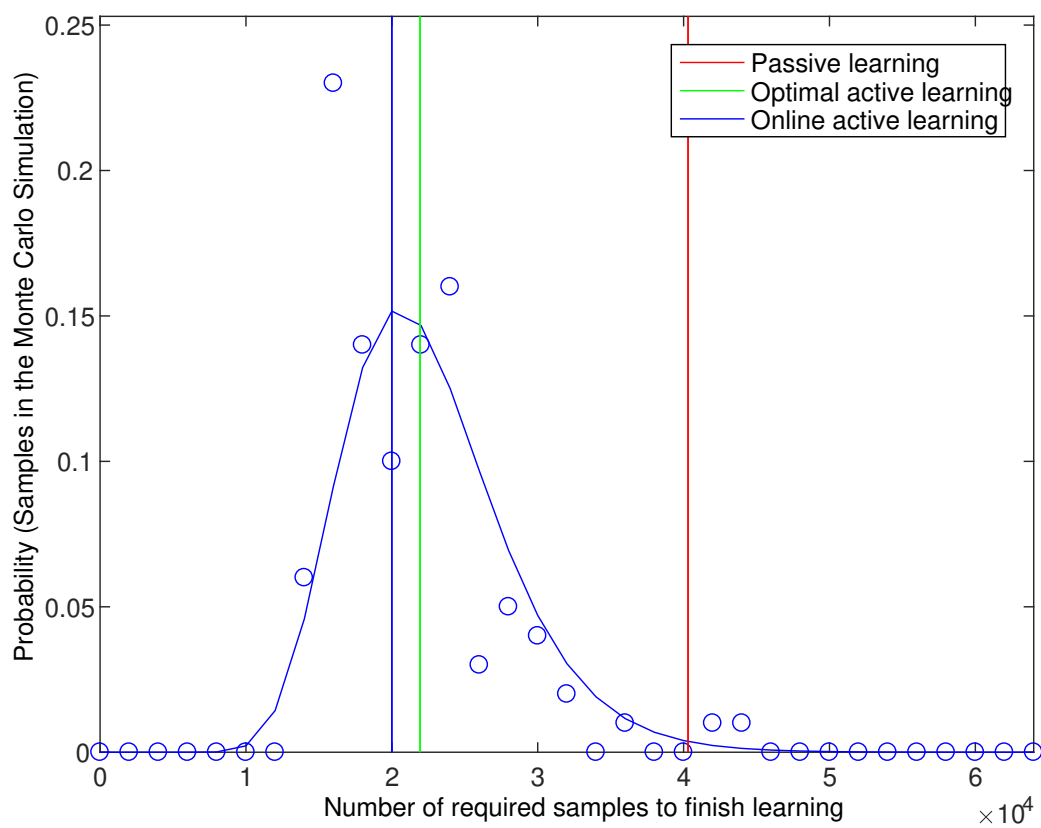


Figure 5. The probability distribution of the number of required samples to finish learning for passive (red), optimal (green), and online active learners (blue). On average, online active learning was as fast as optimal active learning.

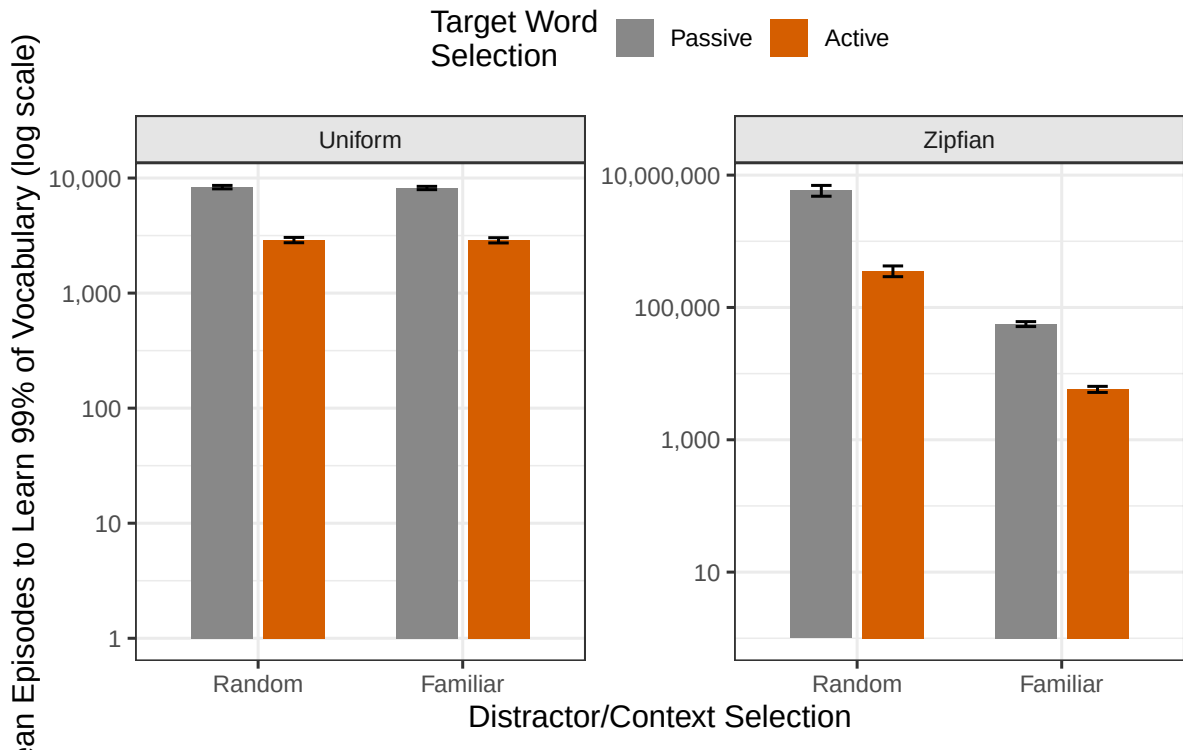


Figure 6. Mean episodes (log scale,  $\pm 1$  SE) required for an incremental, elimination-based learner to acquire 99% of a 1000-word vocabulary, as a function of whether the target word is sampled passively or actively selected to be unknown (bar fill), and whether distractors are drawn randomly or preferentially from familiar (already partly-known) referents (x-axis), separately for uniform and Zipfian word frequency distributions. Results are averaged over context sizes  $C = 10$  and  $C = 100$ .

## Appendix A

The number of samples to complete collection of words given an arbitrary frequency distribution

Here let us derive the number of episodes  $T$  that, for  $0 \leq \epsilon \leq 1$ , the proportion  $(1 - \epsilon)$  of children will have learned all of the  $W$  words listed. Suppose a set of  $W$  words in which each word  $1, \dots, W$  is drawn from the distribution  $p = (p_1, \dots, p_W)$ . The proportion of children who finished learning all the words is  $(1 - \epsilon)$  for  $0 < \epsilon < 1$  requires the number of episodes  $T$ , which is the root of

$$\prod_{i=1}^W (1 - (1 - p_i)^T) = 1 - \epsilon. \quad (4)$$

The left hand side of (4) is the probability that every word is present at least once in the  $T$  episodes.

Write

$$f_{W,\epsilon}(x) := \frac{\log(1 - (1 - \epsilon)^{1/W})}{\log(1 - x)}.$$

For the uniform distribution,  $p_i = 1/W$  for every  $i = 1, \dots, W$ , the root of (4) is given by

$$T = f_{W,\epsilon}(1/W). \quad (5)$$

This  $T$  is the number of episodes with which the proportion of children finished learning all the words is  $(1 - \epsilon)$ . Setting  $\epsilon = 1/2$  in (2), we obtain the median of  $T$ ,  $f_{W,1/2}(1/W)$ , that is comparable with the mean of  $T$  in (1).

Unlike (5) for the uniform distribution, the root  $T$  of Equation (4) in general is not closed-form. Thus, let us consider the upper and lower bound for the root instead of the rigorous form of it. For the general word distribution  $p = (p_1, \dots, p_W)$ , the intermediate value theorem states that there exists a unique constant  $c$  holding  $\min p \leq c \leq \max p$ , with which the root of (4) is expressed as

$$T = f_{W,\epsilon}(c).$$

Equivalently, we have inequality

$$f_{W,\epsilon}(\max p) \leq T \leq f_{W,\epsilon}(\min p) := T_+. \quad (6)$$

As we are interested in the worst possible estimate of  $T$ , this inequality states that the upper bound  $T_+$  of  $T$  is characterized with the probability to sample the least frequent word  $\min p$ .



## Appendix B

The number of samples as a function of the Zipfian exponent

To describe how sensitive the number of samples required to complete the learning to the non-uniformity of the word frequency, we analyze the Zipf distribution with different exponent parameters. Denote the Zipf distribution with the exponent parameter  $a \geq 0$  by  $p = (1^{-a}/H_{W,a}, 2^{-a}/H_{W,a}, \dots, W^{-a}/H_{W,a})$  where  $H_{W,a}$  is the generalized harmonic number  $H_{W,a} = \sum_{i=1}^W i^{-a}$ . It is reduced to the uniform distribution by  $a = 0$ . The larger the exponent  $a$  is, the minimal probability  $\min p$  is smaller. Thus, here we analyze the upper bound  $T_+$  as a function of the exponent parameter  $a$ .

Write  $T_+ = f_{N,\epsilon}(\min p)$ , which gives a reasonable estimate of the upper bound of the root  $T$  of (4). As a function of the exponent  $a$ , we have

$$\frac{\partial \log T_+}{\partial a} = \frac{p_{\min} \left( \frac{\partial H_{N,a}}{\partial a} / H_{N,a} + \log W \right)}{(1 - p_{\min}) \log(1 - p_{\min})}$$

and further we have

$$\frac{\partial^2 \log T_+}{\partial a^2} \geq 0.$$

This implies the  $T_+$  is a super-exponential monotone function of the exponent  $a$ .

## Appendix C

## Iterated linear programming

Before stating the algorithm formally, the intuition behind it is as follows. Start from an arbitrary situation-choice distribution  $q_0$  and identify  $k_1$ , the word-referent pair least likely to be observed under  $q_0$ —the current “weakest link.” Restrict attention to choices of  $q$  that do not make pair  $k_1$  any less likely than it already is, and, among those, pick the  $q$  that maximizes pair  $k_1$ ’s probability; call this  $q_1$ . Now identify the next-weakest pair  $k_2$  under  $q_1$ , add the analogous restriction (do not make  $k_1$  or  $k_2$  any worse), and re-optimize to get  $q_2$ ; repeat. Each iteration locks in a guarantee for one more word-referent pair without loosening the guarantees already made for previously-fixed pairs, so after at most  $N$  iterations, the resulting  $q$  cannot be changed to improve any pair’s probability without decreasing some other pair’s probability that is already at least as small—which is exactly the max-min condition that defines  $\hat{q}$ .

For a  $N \times M$  matrix  $P$ , write its  $i^{\text{th}}$  row by  $P_i$ . Let  $I = \{1, 2, \dots, N\}$  be the set of all indices. At the initial step, define

$$K_0 := \emptyset, \quad C_0 := \mathbb{S}^M, \quad q_0 := e_1,$$

where  $e_1 := (1, 0, \dots, 0)^T \in \mathbb{R}^N$ . Then for  $0 < n \leq N$ , define

$$\begin{aligned} k_n &:= \arg \min_{k \in I \setminus K_{n-1}} P_k q_{n-1}, \quad K_n := K_{n-1} \cup \{k_n\}, \\ C_n &:= \left\{ q \in C_0 \mid \bigwedge_{k \in K_n} (P_{k_n} - P_k)q \leq 0 \right\}, \\ q_n &:= \arg \max_{q \in C_n} P_{k_n} q, \end{aligned}$$

until  $n = m$  such that  $\min_{k \in K_m} P_k q_m \leq \min_{k \in I} P_k q_m$ . The algorithm stops the iterative procedure by outputting  $q := q_m$ .

## Appendix D

Why is online learning possible in the active learning setting?

In this appendix, we describe technical details of the *online* active learning algorithm, which iteratively and asymptotically estimates the optimal scene probability for a given conditional probability matrix  $P \in \mathbb{S}^{N \times M}$

$$\hat{q} = \arg \max_{q \in \mathbb{S}^M} \min(Pq).$$

As shown in our paper, the optimal active learning is exactly computable by iterated linear programming (ILP). This procedure, however, give us little insight about how we can build its “online” version of algorithm due to its discrete nature. Thus, we develop yet another algorithm which also gives near-optimal choice probability  $q$  in a quite different way.

This is based on the gradient of the function  $f := \min(Pq)$ . This is a natural idea at a glance, but the reason why we derived the ILP algorithm rather than the one with the gradient is that the function  $f$  has many non-differentiable subset in its support  $\mathbb{S}^M$ . Especially, in general, for the subset maximizing the function  $f$  is always non-differentiable (with infinitely many tangent plane). Thus, from this theoretical point of view, any algorithm based on the differential may not seem better.

Accordingly, instead of directly working on the function  $f$ , let us consider the family of functions with the parameter  $\lambda$ :

$$g_\lambda(q) := \frac{\sum_i P_i q e^{-\lambda P_i q}}{\sum_i e^{-\lambda P_i q}}$$

where  $P_i$  is the  $i^{\text{th}}$  row of the matrix  $P$ . It is easy to see this function  $g$  has the identity to the function  $f$ :  $\lim_{\lambda \rightarrow \infty} g_\lambda(q) = f(q)$ . Thus, for some sufficiently large  $\lambda$ ,  $g_\lambda(q)$  can approximate  $f(q)$  arbitrarily well, while  $g$  is differentiable everywhere if  $\lambda < \infty$ .

Let us write  $E[h(i)] := \frac{\sum_i h(i) e^{-\lambda P_i q}}{\sum_i e^{-\lambda P_i q}}$  for an arbitrary function  $h(i)$ . The gradient of  $g_\lambda = E[P_i q]$  is

$$\frac{\partial g_\lambda(q)}{\partial q} = E[(1 + \lambda(g_\lambda(q) - P_i q)) P_i^T].$$

With this, in the limit, we have  $\lim_{\lambda \rightarrow \infty} \frac{\partial g_\lambda(q)}{\partial q} = P_{\min}^T$ , where  $P_{\min} := P_{\arg \min_i P_i q}$ .

This gradient is already quite suggestive: the gradient of  $g$  in the limit is a function of both  $q$  and the minimum index  $j = \arg \min_i P_i q$ . Namely, the gradient reflects the tentative hyper-plane  $P_{\min} q = C$ . As long as taking some small step toward this gradient (or with the Lagrange multiplier for a constrained  $q$ ) like the gradient descent method, we can find a minimal  $f(q)$ . Specifically, for  $q \in \mathbb{S}^M$ , set an initial value  $q_0 \in \mathbb{S}^M$  and some small learning rate  $\alpha > 0$ , and for  $n \geq 0$  define

$$q_{n+1} \propto e^{\log q_n + \alpha \text{diag}(q_n)(P_{\min}^T - P_{\min} q_n)}.$$

Then the series of  $q_0, q_1, \dots$  leads to a near-optimal probability. Unfortunately, for the reason mentioned above, this series approaches the optimal solution only up to the scale depending on the learning rate  $\alpha$ , but it has no guarantee to converge to the optimal one. Even with this obvious theoretical flaw of the gradient-based algorithm, it gives us a clear insight about the online algorithm. Replacing the theoretical matrix  $P$  with a sample matrix  $\hat{P}$  which is estimated from a sample, this gradient-based algorithm gives a direct way to build the online substitute for it. Although it still needs to care about the missing values (zero-sample items) in the sample matrix  $\hat{P}$ , it gives a clear picture of why ILP on the sample matrix is informative and possibly gives another online algorithm.